

Predicting Enzyme Class from Protein Sequence

Abstract—As the volume of protein sequences grows significantly, accurate enzyme functional annotation remains a challenge in bioinformatics. This study investigates the classification of enzyme sequences while addressing class imbalance, which is a major issue in biological datasets. Two baseline models, Logistic Regression and XGBoost, were developed using embeddings from the ESM-2 protein language model. The results of the Logistic Regression model showed high recall scores on minority classes, while XGBoost demonstrated high recall scores in majority classes. This finding provided the foundation for a soft-voting ensemble model. The ensemble showed strong performance in feature importance interpretation, demonstrated by heatmaps. Additionally, a confidence estimation was conducted across three levels (High, Medium and Low) to quantify prediction reliability. This study concludes that protein language can effectively capture protein structures, although further research should investigate the incorporation of other features, such as conserved catalytic motif features or predicted active-site residues.

I. INTRODUCTION

Protein sequences that are accessible in the field of bioinformatics are expanding continuously. Examples include the reference protein knowledge base UniProtKB [1], which increased from 180 million available sequences to over 225 million between February 2020 and February 2022. However, the experimental ability to assign function to proteins has not kept pace with the rapid growth of this data [2]. Therefore, the prevalence of unknown functions and misannotations remains a bottleneck in the effort to understand protein function at the molecular level. To bridge the gap between the growing volume of molecular data and the lack of resources available for understanding protein functions, sequence-based and machine learning approaches have been a thriving domain [3].

Enzymes are specialized protein molecules that act as biological catalysts; they play a crucial part in biological processes by speeding up biochemical reactions [4]. Nevertheless, inconsistent nomenclature and complexity have become an issue in naming enzymes. Therefore, the International Union of Biochemistry (IUB) set up the Enzyme Commission to address this challenge by assigning each enzyme a four digit EC number separated by periods, with the first number representing its main class (e.g., EC 3.1.3.16 Protein-serine/threonine phosphatase).

EC prediction is crucial, since this approach facilitates a deeper comprehension of organismal biology and metabolic networks [5]. However, traditional methods used for enzyme classifica-

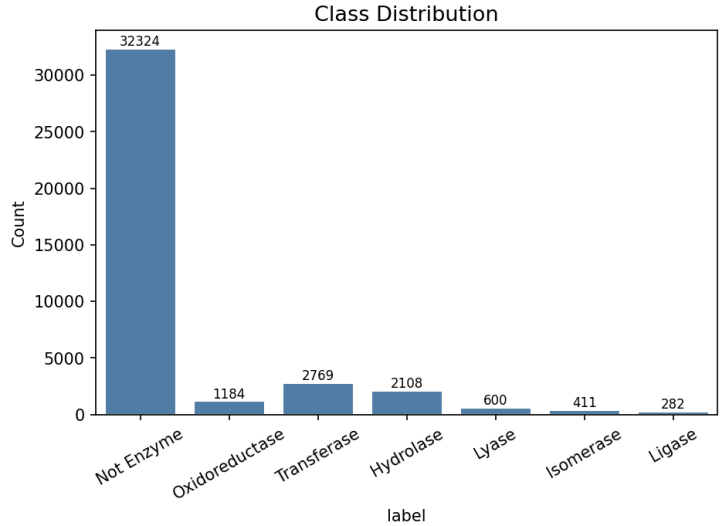


Fig. 1. Class Distribution of the dataset

tion are often computationally expensive and time-consuming. This paper investigates efficient strategies for enzyme classification and proposes solutions to key challenges that are encountered in this field, such as data imbalance.

II. DATA AND METHODS

A. Exploratory Data Analysis

The dataset comprises six enzyme classes and an additional ‘Class 0’ for non-enzyme protein sequences; exploratory analysis revealed a significant class imbalance, with Class 0 dominating the class distribution (Fig. 1).

As shown in Fig. 2, all enzyme classes have similar median sequence lengths. Therefore, sequence length by itself is probably not a reliable diagnostic characteristic for differentiating between different classes of enzymes. However, Classes 2 (Transferases) and 3 (Hydrolases) have a higher amount of long protein sequences, whereas Class 4 (Lyases) has shorter protein sequences compared to those of other classes. Furthermore, Class 0 has one of the lowest median sequence lengths, suggesting that the non-enzyme class consists of smaller proteins that lack the intricate catalytic domains present in bigger enzymes.

B. Data Preprocessing and Sequence Cleaning

Protein sequences underwent a cleaning process before the feature extraction procedure: Strings were converted to uppercase, and whitespace was removed. Non-standard amino acid codes were replaced with the letter ‘X’ to represent unknown residues.

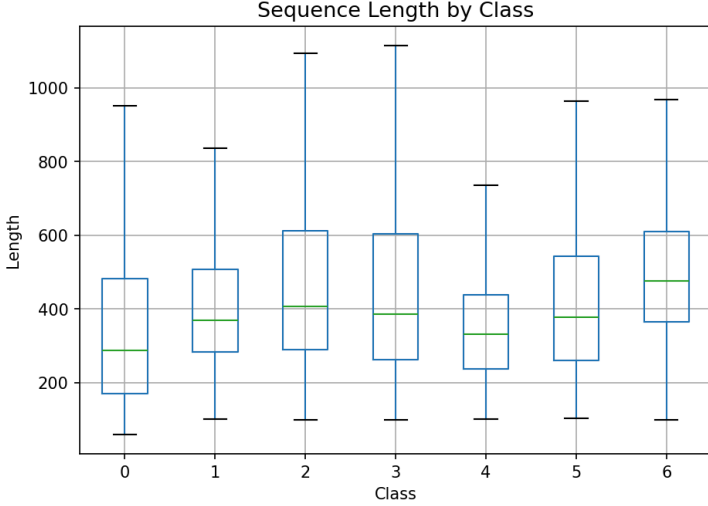


Fig. 2. Sequence lengths of classes

The final dataset had a mean sequence length of 380, a median of 310, and a maximum length of 1200 residues. As the ESM-2 Protein Language Model requires a hard limit of 1022 residues, sequences longer than this length were truncated.

C. Feature Extraction

a) *Protein Language Model Embeddings (ESM-2)*: To derive sequence representations, the transformer-based protein language model (PLM) ESM-2 was used. ESM-2 is a powerful model for capturing the function and structure of proteins, and is commonly used for tasks such as protein function annotation [6].

A relatively small ESM-2 model, consisting of 6 layers and 8M parameters (esm2_t6_8M_UR50D) was used. During this procedure, mean-pooling was performed by averaging the representations extracted from the final layer of the model.

D. Handling Class Imbalance

With Class 0 (Non-enzyme) containing 32,410 sequences compared to Class 6 (Ligases) with 282 sequences, the dataset showed an extreme class imbalance, with a ratio of approximately 115:1. To address this issue, some precautions were taken during model initialisation and training.

Initially, during the training of the Logistic Regression and XGBoost models, class-balanced sample weighting was applied, where each class was assigned a weight inversely proportional to their sample size.

Secondly, evaluation metrics like the macro-averaged F1 score, balanced accuracy, and Matthews Correlation Coefficient (MCC), were purposefully selected to represent class performance equally, regardless of class sample size.

Third, a soft-voting ensemble was used to complement the strengths of the two base models. Logistic Regression obtained

higher recall on minority classes, whereas XGBoost performed better on majority classes, as explained in Section 2.5.

E. Model Architectures

Two baseline models, Logistic Regression and XGBoost, were developed for this project. Both models were combined for soft-voting.

a) *Logistic Regression Baseline*: The logistic regression with the multinomial formulation was chosen as the baseline primarily due to its simplicity, computational efficiency, and its ability to handle imbalanced class problems. Hyperparameter tuning was applied between $C = \{0.01, 0.1, 1, 10\}$ to find the optimal value of C depending on the F1 macro value. $C = 0.01$ was selected. Class-balanced sample weighting was applied, leading the weights of the imbalanced class to be increased.

The model showed higher recall for minority classes (4, 5 and 6) than XGBoost classifier, meaning that ESM-2 embedding space is somewhat linearly separable. Additionally, linear classifiers might perform better than non-linear classifiers, especially in cases of class imbalances.

b) *XGBoost Classifier*: XGBoost (Extreme Gradient Boosting) was selected as the primary model due to its efficiency and speed. Furthermore, the model was configured to use a subsample of 80% of the training data and 80% of the features per tree. Class imbalance was again addressed.

The model achieved high recall in the majority classes (0, 2 and 3) but a lower recall score in the minority classes, which motivated the use of the soft-voting ensemble.

F. Experimental Design and Cross-Validation

5-fold cross-validation was used throughout the model development and evaluation. To account for class imbalance, stratified sampling was applied in each fold.

G. Evaluation Metrics

In addition to accuracy, various evaluation metrics were used during the model assessment. Firstly, macro-averaged F1 score was used as the primary metric because it treats all classes equally. Balanced accuracy, which computes the average recall per class, was included as a reliable measure of model performance. Additionally, Matthews Correlation Coefficient (MCC) was also selected because it incorporates all four values from the confusion matrix (TP, TN, FP, and FN), giving an insightful evaluation when there is a class imbalance

H. Confidence Estimation and Calibration

a) Confidence Score Computation:

Each prediction of the final ensemble model was assigned a confidence estimate to quantify the classification certainty. This confidence score was derived from the maximum predicted

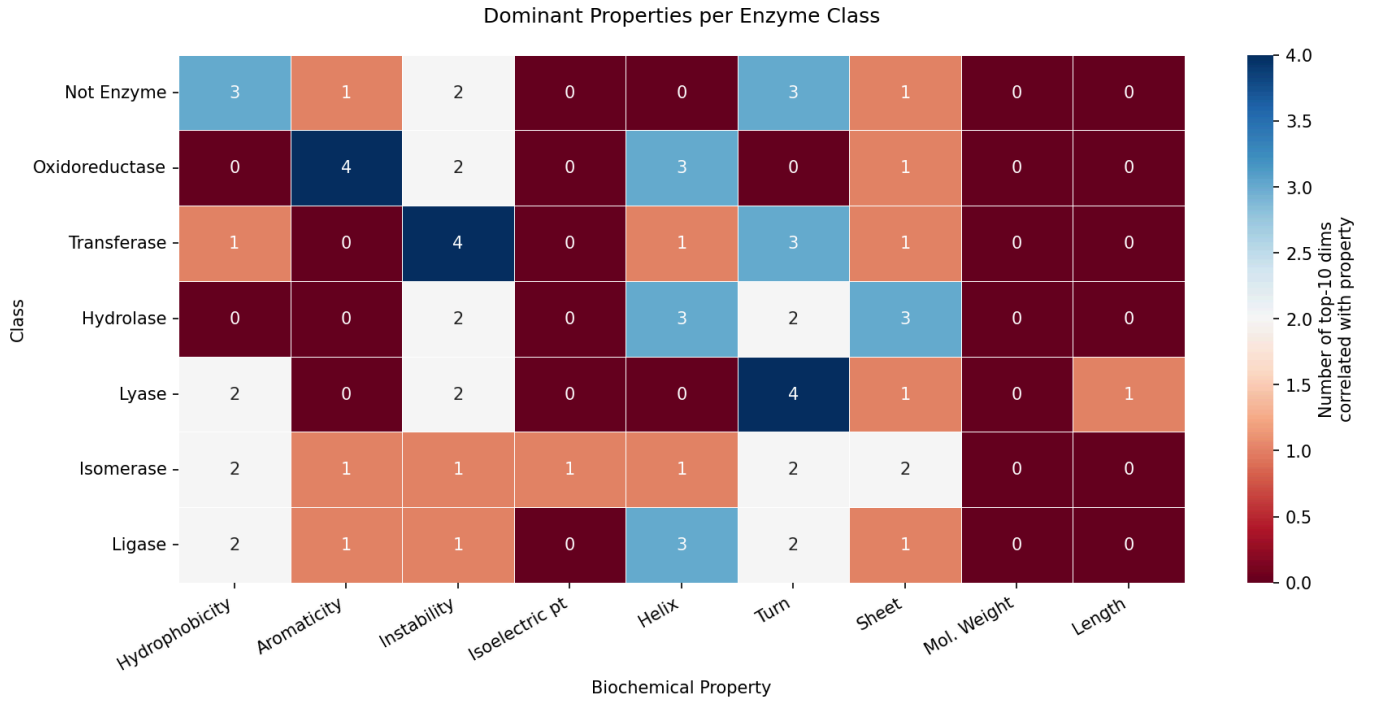


Fig. 3. Dominant Properties per Class

probability across all seven classes, which corresponds to the probability assigned to the predicted class, and it ranged from 0.14 (1/7) to 1, for any particular prediction.

This score was then assigned to three different levels:

- **High:** confidence score > 0.8
- **Medium:** confidence score between 0.5 and 0.8
- **Low:** confidence score < 0.5

III. RESULTS

A. Interpretability Analysis

To explore the most dominant property of each enzyme class, a heatmap was produced (Fig. 3).

- **Non-Enzyme:** It can be concluded that non-enzyme sequences were classified based on hydrophobicity and turn-fraction signals
- **Oxidoreductase:** The most dominant property for this class was aromaticity, confirming its role in binding aromatic cofactors
- **Transferase:** This class exhibited the instability index as its most prominent property. Since they are involved regulatory signaling pathways, their function requires flexibility and instability
- **Hydrolase:** This class showed equal dominance in sheet and helix content, reflecting them sharing α - β fold structures
- **Lyase :** The most outstanding property of this class was its turn fraction which relates to its complex structure and loop-mediated active sites

- **Isomerase :** This class did not show any notable properties, with all features contributing nearly equally. This finding, alongside the insufficient sample size, might explain the poor classification performance
- **Ligase :** The helix fraction dominated this class, explaining their presence of α -helical bundles

B. Cross-Validation Performance Across Models

All models underwent five-fold stratified cross-validation on the training set, and the findings can be seen in Table I.

TABLE I. CROSS VALIDATION PERFORMANCE ACROSS ALL MODELS

Model	Accuracy	Macro F1	Balanced Acc	MCC
Logistic Regression	0.7215	0.4133	0.6037	0.4496
XGBoost	0.8368	0.5086	0.5246	0.5529
Ensemble [LR=1, XGB=2]	0.8679	0.5062	0.4538	0.5510
Ensemble [LR=2, XGB=1]	0.8350	0.4949	0.5731	0.5445

Logistic Regression was the model that achieved the lowest accuracy and Macro F1 score, at approximately 72% and 41%, respectively. However, despite its lower accuracy, the model achieved high recall on the three minority classes (Lyase (0.5),

Isomerase (0.39), and Ligase (0.74)) in comparison to XGBoost (See Appendix A).

On the other hand, XGBoost achieved a higher accuracy and Macro F1 score, at approximately 83% and 50%, respectively. Its performance on the majority classes was higher, especially for Classes 0, 2 and 3 (See Appendix A). However, despite the use of balanced sample weights, the model showed lower recall on the minority classes, indicating that its boosting mechanism still favored more prevalent class samples.

Two soft-voting ensembles were configured to leverage both models’ strengths. Assigning a higher weight to XGBoost resulted in the highest accuracy and Macro F1. However, this also led to the lowest recall scores among the minority classes. On the contrary, when the weights were adjusted in favor of Logistic Regression, the findings show more balanced accuracy and improved recall across the three minority classes.

The [LR=2, XGB=1] model was chosen as the final model due to its ability to provide a balance between accuracy and sensitivity, despite the precision-recall tradeoff.

C. Final Held-Out Test Set Evaluation

The selected ensemble model was trained on the model and tested on 7,936 sequences. Table II shows the comparison of the validation performance and cross-validation data.

TABLE II. CROSS-VALIDATION AND FINAL TEST SET PERFORMANCE OF THE ENSEMBLE MODEL [LR=2, XGB=1]

Split	Accuracy	Macro F1	Balanced Acc	MCC
Cross-validation (mean)	0.8350	0.4949	0.5731	0.5445
Test	0.8328	0.4916	0.5846	0.5416
Difference	-0.0022	-0.0033	+0.0115	-0.0029

The small discrepancy between the cross-validation and test set results indicates that there was no overfitting. As can be seen from the Fig. 4, performance in classification differed significantly among the seven classes. Majority classes achieved high precision score, with Class 0 obtaining the highest precision, recall and F1 scores. Class 2 (Transferases) achieved the highest F1-score of 0.63 (See Appendix B).

The three minority classes had the lowest overall scores. A significant finding is that Ligases achieved a high recall score of 0.84, despite having a 0.25 precision score. This indicates a high false-positive rate, where sequences from other class were often misclassified as Ligases.

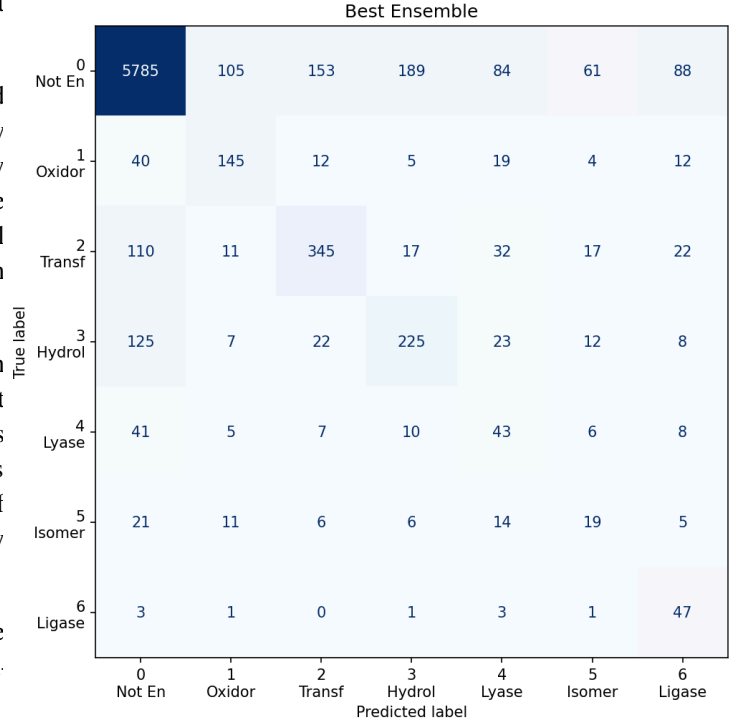


Fig. 4. Test set confusion matrix

The confusion matrix showed that the misclassification of enzyme sequences as non-enzymes (Class 0) was the primary cause of classification error, with 680 true enzyme sequences in the test set mistakenly assigned to Class 0. This suggests that defining the decision boundary between enzymatic and non-enzymatic sequences continues to be a major bottleneck. The most common enzyme-to-enzyme misclassification occurred between Hydrolase and Transferase; which have structural and functional characteristics that might lead to overlapping ESM-2 representations. Otherwise, inter-class misclassification between enzyme classes was rather low.

D. Confidence Calibration

To validate confidence estimation, the held-out test set predictions were divided into three groups based on the given confidence level (High, Medium and Low), and each group’s classification accuracy was calculated separately.

The thresholds for each level were selected based on the output probability distribution. A ‘Gold Standard’ subset indicated a ‘High’ confidence, predictions that failed to achieve a majority agreement were isolated under the ‘Low’ threshold, indicating the need for further attention.

As demonstrated in Table III, high confidence predictions which were 41.7% of the test set achieved a 97.4% accuracy, offering a reliable prediction certainty. Conversely, low confidence predictions achieved an overall 55.7% accuracy, which is lower than the model accuracy. Only 22.9% of the test set was

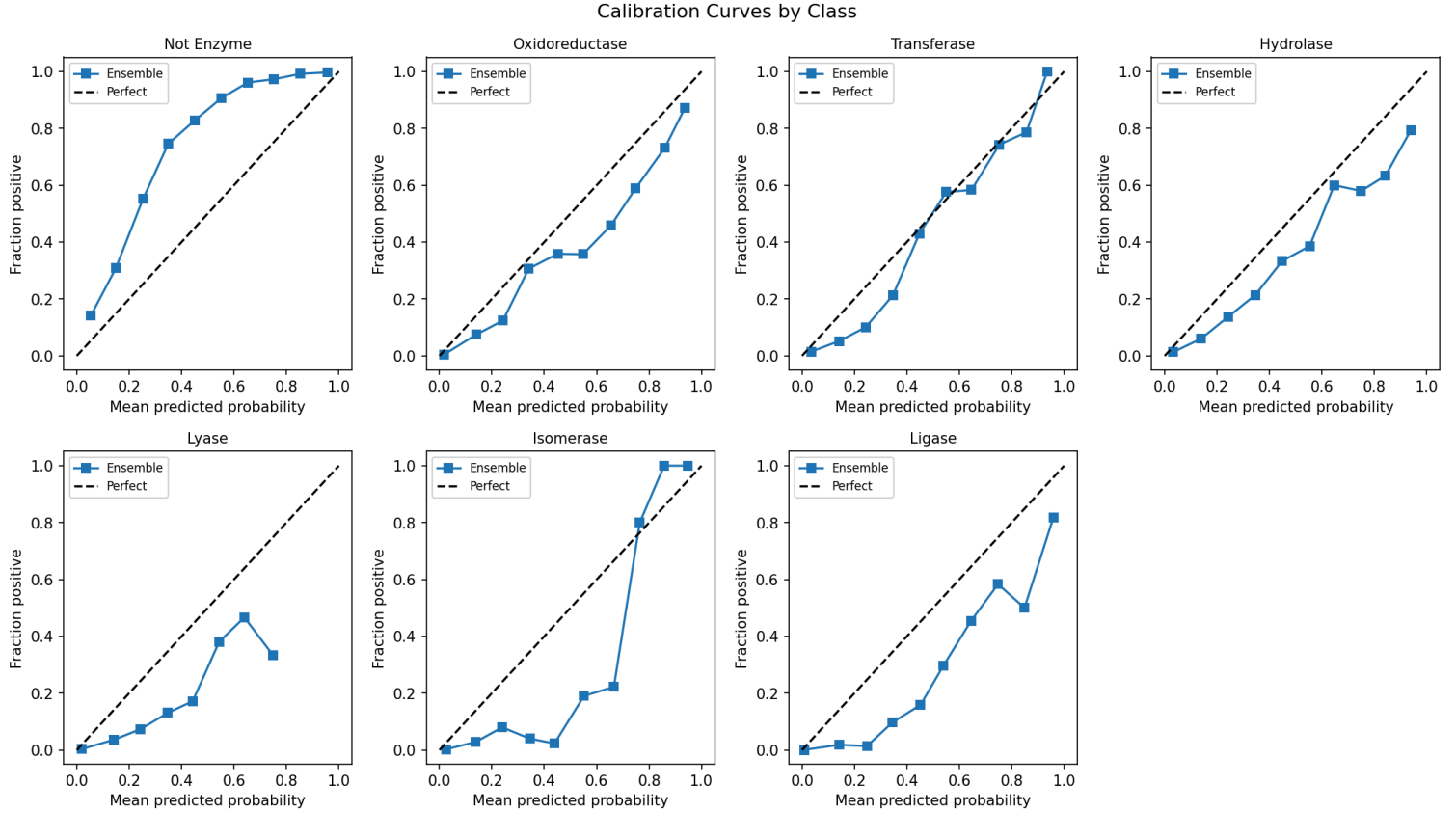


Fig. 5. Calibration Curves by Class

TABLE III. CONFIDENCE LEVEL DISTRIBUTION AND ACCURACY ON THE HELD-OUT TEST SET

Confidence Level	Threshold	n	% of Test Set	Accuracy
High	≥ 0.80	3,124	41.7%	0.974
Medium	$0.50 - 0.80$	2,809	35.4%	0.844
Low	< 0.50	1,815	22.9%	0.557

classified as low confidence, suggesting that these sequences require further analysis (See Appendix D for blind predictions). The calibration curve shown in Fig. 5 indicates that due to the weak calibration of minority classes, such as Lyases and Isomerases, confidence scores for these predictions should be treated cautiously.

IV. DISCUSSION

A. Class Imbalance and Its Effect on Minority Classes

With the non-enzyme class dominating the dataset and the smallest enzyme class representing less than 0.4% of sequences, class imbalance was the primary challenge in this study. Despite the implementation of balanced sample weighting during training and a soft-voting ensemble designed to

preserve sensitivity of the minority classes, these classes still demonstrated low F1-scores.

One way to address this challenge is through the SMOTE method, which was explored during the study but was excluded from the pipeline due to computational limitations.

B. Distinguishing Enzyme Classes from Non-Enzymatic Proteins

Enzyme and non-enzyme proteins share common structural cores [7]. This observation is further supported by the confidence matrix in Fig. 4, where the main source of error was the enzymes being classified as non-enzymes. Arguably, enzyme vs non-enzyme classification is more difficult than inter-class enzyme classification, as the locations of these protein sequences within the ESM-2 embedding space might be overlapping. This highly complex decision boundary could be improved in future work by implementing predicted active site residues.

C. Limitations and Future Directions

Some limitations in this project include the choice of the ESM-2 model. Firstly, the smallest model (esm2_t6_8M_UR50D) was used to account for memory constraints. Using larger ESM-2 models with more layers and parameters can improve the classification of minority classes.

Additionally, despite there a small portion of protein sequences exceeding 1022 residues, the truncation of these sequences might have caused a loss of crucial information from the C-terminal.

Furthermore, this study did not implement conserved motifs associated with catalytic activity or predicted active-site residues, both of which have been suggested as highly informative features for this classification task. Future research could incorporate these features to improve the classification performance of minority classes.

Finally, the weights for the final model were selected from only three configurations, and the XGBoost model did not undergo a hyperparameter tuning process. Further research could include a more extensive optimisation of these parameters to improve the overall performance.

APPENDIX

Appendix A: Per-class precision, recall, and F1 score for all models under five-fold cross-validation

TABLE IV. LOGISTIC REGRESSION WITH ESM

Class	Precision	Recall	F1-Score	Support
Not Enzyme	0.97	0.75	0.85	25,859
Oxidoreductase	0.39	0.65	0.49	947
Transferase	0.47	0.61	0.53	2,215
Hydrolase	0.36	0.58	0.45	1,686
Lyase	0.12	0.50	0.19	480
Isomerase	0.08	0.39	0.14	329
Ligase	0.15	0.74	0.25	226
Accuracy			0.72	31,742
Macro Avg	0.36	0.60	0.41	31,742
Weighted Avg	0.86	0.72	0.77	31,742

TABLE V. XGBOOST WITH ESM

Class	Precision	Recall	F1-Score	Support
Not Enzyme	0.95	0.90	0.92	25,859
Oxidoreductase	0.51	0.63	0.56	947
Transferase	0.49	0.67	0.57	2,215
Hydrolase	0.45	0.62	0.52	1,686
Lyase	0.23	0.24	0.23	480
Isomerase	0.43	0.19	0.26	329
Ligase	0.58	0.43	0.49	226
Accuracy			0.84	31,742
Macro Avg	0.52	0.52	0.51	31,742
Weighted Avg	0.86	0.84	0.84	31,742

TABLE VI. ENSEMBLE [LR=1, XGB=2]

Class	Precision	Recall	F1-Score	Support
Not Enzyme	0.91	0.96	0.93	25,859
Oxidoreductase	0.62	0.55	0.58	947
Transferase	0.66	0.52	0.58	2,215
Hydrolase	0.63	0.47	0.54	1,686
Lyase	0.31	0.17	0.22	480
Isomerase	0.51	0.14	0.22	329
Ligase	0.64	0.37	0.47	226
Accuracy			0.87	31,742
Macro Avg	0.61	0.45	0.51	31,742
Weighted Avg	0.85	0.87	0.86	31,742

TABLE VII. ENSEMBLE [LR=2, XGB=1]

Class	Precision	Recall	F1-Score	Support
Not Enzyme	0.95	0.90	0.92	25,859
Oxidoreductase	0.51	0.65	0.57	947
Transferase	0.62	0.59	0.60	2,215
Hydrolase	0.50	0.55	0.53	1,686
Lyase	0.22	0.37	0.27	480
Isomerase	0.18	0.29	0.22	329
Ligase	0.24	0.67	0.35	226
Accuracy			0.84	31,742
Macro Avg	0.46	0.57	0.49	31,742
Weighted Avg	0.86	0.84	0.85	31,742

Appendix B: Per-class precision, recall, and F1 score for the final model under the test set

TABLE VIII. VALIDATION PERFORMANCE OF THE FINAL ENSEMBLE MODEL

Class	Precision	Recall	F1-Score	Support
Not Enzyme	0.94	0.89	0.92	6,465
Oxidoreductase	0.51	0.61	0.56	237
Transferase	0.63	0.62	0.63	554
Hydrolase	0.50	0.53	0.51	422
Lyase	0.20	0.36	0.25	120
Isomerase	0.16	0.23	0.19	82
Ligase	0.25	0.84	0.38	56
Accuracy			0.83	7,936
Macro Avg	0.46	0.58	0.49	7,936
Weighted Avg	0.86	0.83	0.84	7,936

Appendix C: Code and Dependencies

The classification procedure was conducted in Python 3.10 using Google Colab with a T4 GPU. The dependencies are: BioPython 1.81, scikit-learn 1.3 (LR, RF, VotingClassifier, cross-validation), XGBoost 2.0 (gradient boosting), imbalanced-learn (SMOTE), as well as standard scientific libraries (NumPy, pandas, matplotlib, seaborn). Code design include:

(1) Sequence cleaning, (2) ESM-2 embedding extraction, (3) Stratified train-test split, (4) Cross-validation, (5) LR and XGBoost implementation, (6) Ensemble implementation, (7) Interpretability analysis

Appendix C: Blind Predictions

SEQ01 2 Confidence Medium
SEQ02 3 Confidence Low
SEQ03 3 Confidence Medium
SEQ04 1 Confidence Medium
SEQ05 3 Confidence High
SEQ06 0 Confidence Low
SEQ07 3 Confidence High
SEQ08 2 Confidence Medium
SEQ09 2 Confidence Low
SEQ10 1 Confidence Medium
SEQ11 2 Confidence High
SEQ12 2 Confidence Low
SEQ13 4 Confidence Low
SEQ14 1 Confidence High
SEQ15 2 Confidence Medium
SEQ16 4 Confidence Low
SEQ17 2 Confidence High
SEQ18 2 Confidence High
SEQ19 2 Confidence Medium
SEQ20 2 Confidence Low
SEQ21 3 Confidence Low
SEQ22 0 Confidence High

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